

Childcare siting — results

Emma Ventresca · 2026-10-08

Question 1 — ACS 5-year explanatory variables, county level, 2015–2022

Process completed. See AI output notes below.

Question 2 — OEWS childcare-worker wages, SOC 39-9011, area level, 2015–2022

Process completed. See AI output notes below.

Question 3 — Merge and cleaning

Process completed. See AI output notes below.

Year	Rows	Missing wage	% missing
2015	1,233	11	0.89
2016	1,233	14	1.14
2017	1,234	10	0.81
2018	1,233	5	0.41
2019	1,235	6	0.49
2020	1,235	20	1.62
2021	1,235	13	1.05
2022	1,235	20	1.62
Total	9,873	99	1.00

Question 4 — Descriptive statistics, 2022

Predictor	Mean	Std. dev.	N
Per capita income (\$)	36,156.1	10,504.1	1,228
Total population	231,862.6	506,835.2	1,228
Own children under 6	15,440.9	34,275.0	1,228
Own children <6, all parents in LF	10,419.9	22,637.0	1,228
Median gross rent (\$/mo)	1,051.8	341.9	1,227
Share women 25+ bachelor's+	0.3013	0.1070	1,228
Share women 16–64 FT year-round	0.4367	0.0683	1,228
Share of population under 6	0.0660	0.0124	1,228
Share kids under 6 needing care	0.6728	0.0906	1,227
Childcare worker mean annual wage (\$)	27,373.5	4,337.1	1,215

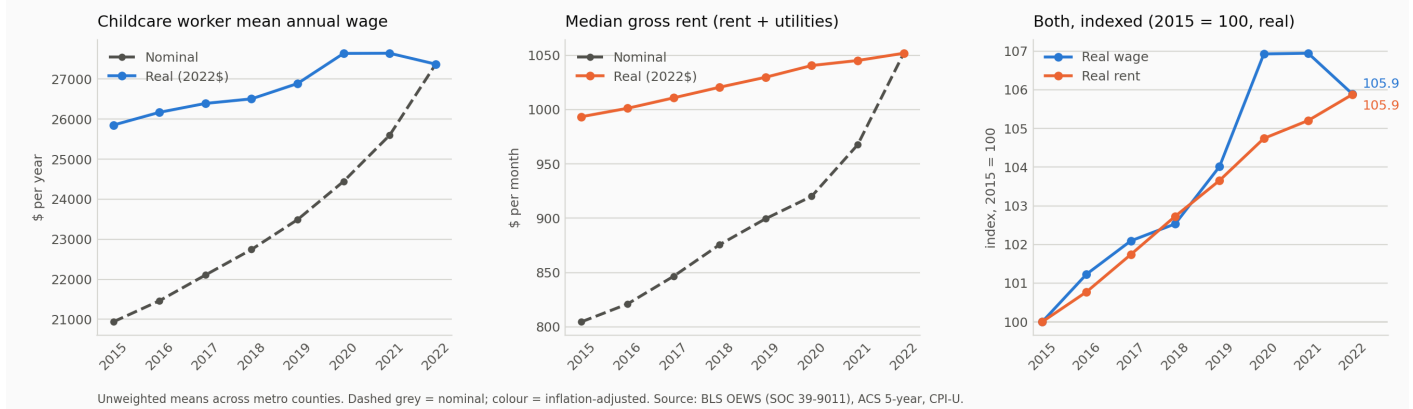
Question 5 — Constant 2022 dollars (CPI-U)

$\text{real_2022} = \text{nominal_t} \times (\text{CPI_2022} / \text{CPI_t})$; CPI-U series CUUR0000SA0.

Year	CPI-U	Deflator	Wage nominal (\$/yr)	Wage real (\$/yr)	Rent nominal (\$/mo)	Rent real (\$/mo)
2015	237.017	1.2347	20,937	25,851	805	993
2016	240.007	1.2194	21,462	26,170	821	1,001
2017	245.120	1.1939	22,106	26,393	847	1,011
2018	251.107	1.1655	22,743	26,506	876	1,020
2019	255.657	1.1447	23,490	26,889	899	1,030
2020	258.811	1.1308	24,445	27,641	920	1,040
2021	270.970	1.0800	25,598	27,646	968	1,045
2022	292.655	1.0000	27,373	27,373	1,052	1,052

Change 2015→2022	Nominal	Real
Childcare wage	+30.75%	+5.89%
Median gross rent	+30.73%	+5.87%

Childcare worker wages and rents in OEWS metropolitan counties, 2015–2022 (constant 2022 dollars, CPI-U)



Question 6 — Mean real wage by year, and 2019 vs 2022

$\text{SE} = \text{sd} / \sqrt{n}$; sd is the sample standard deviation (divisor $n-1$).

County level

Year	N counties	Mean real wage (\$2022)	Std. dev.	Std. error
2015	1,222	25,851.21	3,266.89	93.4543
2016	1,219	26,169.71	3,371.54	96.5665
2017	1,224	26,392.52	3,514.92	100.4673
2018	1,228	26,505.90	3,702.79	105.6648
2019	1,229	26,888.87	3,897.24	111.1685
2020	1,215	27,641.46	4,423.70	126.9104
2021	1,222	27,646.21	4,464.55	127.7151
2022	1,215	27,373.48	4,337.10	124.4261

OEWS area level

Year	N areas	Mean real wage (\$2022)	Std. dev.	Std. error
2015	381	25,937.65	3,270.72	167.5641
2016	381	26,278.38	3,323.28	170.2566
2017	384	26,510.27	3,475.07	177.3366
2018	385	26,710.36	3,727.65	189.9789
2019	385	27,189.98	4,021.61	204.9603
2020	376	28,126.27	4,579.23	236.1560
2021	382	28,041.27	4,553.03	232.9533
2022	377	27,716.31	4,409.76	227.1140

H_0 : mean real wage 2019 = mean real wage 2022 (two-tailed)

Specification	diff (2022–2019)	SE(diff)	t	df	p	95% CI
Two-sample, county level	484.6115	166.8541	2.9044	2,408.5	0.00371305	[157.4191, 811.8039]
Two-sample, OEWS area level	526.3327	305.9240	1.7205	750.4	0.0857595	[-74.2360, 1,126.9014]
Paired, county level (n = 1,212)	456.3581	53.2066	8.5771	1,211	2.93259e-17	[351.9708, 560.7455]
Paired, OEWS area level (n = 374)	497.1583	111.5536	4.4567	373	1.1023e-05	[277.8056, 716.5111]

Paired SE = $\text{sd}(x_{2022} - x_{2019}) / \text{sqrt}(n)$.

Question 7 — Childcare wage on female college share, 2022

$a_mean_i = b_0 + b_1 \cdot \text{share_women_25plus_bachelors_plus_i} + e_i$ (OLS, 2022)

Term	Coefficient	Std. error	t	P> t	95% CI
share_women_25plus_bachelors_plus (b1)	16,499.9862	1,058.1894	15.5927	4.48293e-50	[14,423.8894, 18,576.0829]
Intercept (b0)	22,359.1942	339.1614	65.9249	0	[21,693.7823, 23,024.6060]

Statistic	Value
N	1,208
R ²	0.167777
Adjusted R ²	0.167087
Root MSE	3,949.9201
F(1, 1,206)	243.131
p(F)	4.48293e-50
Residual df	1,206

Standard errors clustered by OEWS area (373 clusters):

Term	Coefficient	Cluster-robust SE	t	P> t
share_women_25plus_bachelors_plus (b1)	16,499.9862	2,217.3295	7.4414	6.97122e-13
Intercept (b0)	22,359.1942	606.3146	36.8772	2.65e-126

AI output notes

AI Prompt Log

Student: Emma Ventresca **Assignment:** Childcare siting — county-level data build and results, 2015–2022 **AI tool used:** Claude (Opus 5) via Claude Code **Session date:** 2026-10-08

This file is a disclosure record. Every prompt given to the AI in this session is reproduced below word-for-word, in order, followed by exactly what the AI did in response. Where a statistical procedure was run, its specification and output are recorded with the prompt.

Problem 1 — ACS 5-year estimates (county-level explanatory variables)

Prompt 1

Hey, okay, under the homework five folder, can you set up a similar AI prompt log? And first thing we're gonna do is I have a census data API key. Tell me where to put that so it's safe and doesn't get like thrown into uh, some kind of like public repo or something like that or in this write-up. And here's what we're gonna do. You are going to go onto the census website with my key and you are going to download um, the ACS five-year estimates for the explanatory variables uh, that I'm going to give you. And this needs to be at the county level from 2015 through 2022 inclusive. And here are the variables that we need. We need per capita income, total population, own children under six, under six with all resident parents and labor force, median gross rent, which is rent plus utilities, share women 25 plus with bachelor's plus, share women 16 to 64 full-time year-round. So that is eight different categories. And we also need the following. Those refer demands and costs. Now we need some young child data. We need three categories, which are share kids under six, share kids under six who need care. Actually, that's just two categories, so just do those. Okay, go do that and tell me where I can put in my API key.

Actions taken by AI:

- Confirmed API-key safety before anything else.** Verified by direct check that the project `.env` is untracked by git, is matched by `.gitignore` (`.env`), and has never appeared in the repository history. The key is pasted into `.env` only.
- Verified every Census variable ID against the live API** before writing any code, by requesting each variable's metadata from `api.census.gov/data/2022/acs/acs5/variables/<id>.json` and reading back its official label. All 18 IDs confirmed. No variable was chosen from memory.
- Determined a key is genuinely required.** A keyless request to the ACS *data* endpoint returned HTTP 302 and redirected to an HTML documentation page rather than JSON. (The variable-*metadata* endpoint does serve keyless, which is how step 2

was done without the key.)

4. **Wrote `scripts/12_fetch_acs.py`** . Reads the key from `CENSUS_API_KEY` (environment or `.env`), never prints it, never writes it to any output, and logs request URLs with the key redacted. Saves each year's API response **verbatim** to `data/raw/acs/acs5_county_<year>.json` and writes derived values to a separate file, so no raw data is edited in place.
5. Ran the script to confirm it fails safely with no key present (exit 1, writes nothing).

(Recorded decision: asked which folder to use, since `Stats 50M/hw5` holds a different assignment. The user selected `final_project_stats` , whose `README` already listed this exact ACS pull as outstanding.)

Variables requested, and the ACS table/variable each maps to (ACS 5-year, county level, 2015–2022 inclusive):

Demand and cost

#	Requested	ACS variable(s)	Official label
1	Per capita income	B19301_001E	Per capita income in the past 12 months
2	Total population	B01003_001E	Total population
3	Own children under 6	B23008_002E	Total: Under 6 years
4	Under 6 w/ all resident parents in labor force	B23008_004E + B23008_010E + B23008_013E	Two parents both in LF; one parent (father) in LF; one parent (mother) in LF
5	Median gross rent (rent + utilities)	B25064_001E	Median gross rent — gross rent is contract rent plus utilities by Census definition
6	Share women 25+ with bachelor's+	(B15002_032E + _033E + _034E + _035E) / B15002_019E	Female bachelor's / master's / professional / doctorate, over female 25+ total
7	Share women 16–64 full-time year-round	B23022_029E / B23022_026E	Female, usually 35+ hrs/wk, 50–52 weeks, over female 16–64 total

Young-child shares

#	Requested	Construction
8	Share kids under 6	(B09001_003E + B09001_004E + B09001_005E) / B01003_001E
9	Share kids under 6 who need care	(B23008_004E + B23008_010E + B23008_013E) / B23008_002E

Both shares are **derived**, not ACS variables in their own right. The script writes the underlying numerator and denominator counts alongside each share so every one can be checked by hand.

(The prompt named seven distinct demand/cost series but called them eight; the user confirmed seven in Prompt 2.)

Notes affecting later analysis: - ACS dollar figures are published in each year's own inflation-adjusted dollars and still require CPI-U rebasing to constant 2022 dollars (done in Problem 5). - `county_fips` is written as a 5-character zero-padded **string**. - ACS missing/suppressed sentinels (`-66666666` and similar) are converted to blank rather than carried through as large negative numbers.

No data was downloaded at this point — awaiting the API key. **No regression or statistical test was run.**

Problem 2 — OEWS childcare-worker wages (SOC 39-9011)

Prompt 2

your right only 7 and yes constrcut it yourself and list this all under problem 1. for probelm two you are gonna use the ble owes to donunlaod childcare-worker wage earnings at the area level , 2015-2022 and this is SOC 39-9011, extrac those

Actions taken by AI:

1. **Recorded two confirmations:** the demand/cost list is seven series, not eight; and the AI's construction of "share kids under 6 who need care" is approved.
2. **Downloaded the OEWS area files for 2015–2022** from <https://www.bls.gov/oes/special-requests/oesm<YY>ma.zip> — eight archives, ~287 MB total, saved unmodified to `data/raw/oews/`. No API key required.
3. **Extracted SOC 39-9011 (Childcare Workers)** from both workbooks in each archive: `MSA_M<year>_dl.xlsx` (metropolitan) and `BOS_M<year>_dl.xlsx` (nonmetropolitan "balance of state"). Both are needed because the project crosswalk contains areas of both kinds. The `aMSA_M<year>_dl.xlsx` file present in 2015–2017 is an alphabetical re-sort of the MSA file and was skipped to avoid double-counting.
4. **Normalised column names across years.** BLS changed these mid-period: 2015–2018 use uppercase `AREA / AREA_NAME`; 2019 is entirely lowercase; 2020–2022 are uppercase with `AREA_TITLE`. All folded to lowercase with `area_name` mapped to `area_title`.
5. **Wrote `scripts/13_fetch_oews.py`** and `data/processed/oews_childcare_2015_2022.csv` — 4,328 area-year rows.

Bug found and fixed during this session (disclosed for transparency): the first version of the extraction script produced a file in which almost every `oews_area`, `area_title` and wage value was blank. Cause: after filtering the workbook to SOC 39-9011 the pandas DataFrame retained its original row index, so assigning those columns into a newly built frame aligned on index and silently produced `NaN` instead of raising. Detected by two sanity checks built into the script — a duplicate `(year, oews_area)` count (4,304 of 4,328) and a coverage check against the project crosswalk (16 of 4,271 matched). Fixed with `reset_index(drop=True)` and re-run. **The bad file was overwritten by the corrected run and was never used in any analysis.**

Suppression handling. BLS uses `*` (estimate not released), `**` (data not available) and `#` (wage at or above \$100/hour or \$208,000/year). These are written as blanks in the numeric columns, with the original marker preserved in `a_mean_flag` / `a_median_flag` so nothing is silently coerced to a number.

Mean annual wage, SOC 39-9011, metropolitan areas (nominal, not yet CPI-rebased):

Year	Areas	Mean	Median	Min	Max
2015	414	21,291	20,835	16,900	36,810
2016	414	21,845	21,300	17,140	38,180
2017	417	22,501	22,070	17,460	37,100
2018	392	22,973	22,620	17,680	34,570
2019	392	23,813	23,330	17,620	35,590
2020	383	24,944	24,230	17,330	38,590
2021	389	26,033	25,150	18,320	39,390
2022	384	27,807	26,875	19,540	41,700

Coverage gap found. 116 of 4,271 crosswalk area-years have no OEWS 39-9011 row. In 2015–2017 OEWS published the eleven largest metros **only** as metropolitan divisions (NECTA divisions for Boston), not as the combined MSA the crosswalk uses — e.g. the crosswalk maps Cook County to `0016980` (Chicago-Naperville-Elgin) but the 2015 file contains only `0016974` (Chicago-Naperville-Arlington Heights Metropolitan Division). Affected 2015–2017: New York, Los Angeles, Chicago, Dallas–Fort Worth, Detroit, Miami, Philadelphia, San Francisco, Seattle, Washington DC, Boston. The remaining 83 area-years are small MSAs, Puerto Rico areas and nonmetro areas where 39-9011 was not published. Full list in `docs/oews_unmatched_areas.csv`. This was flagged to the user and resolved in Problem 3.

No regression or statistical test was run.

Key handling

Prompt 3

wait where do i put my key

Actions taken by AI: Displayed the three lines of `.env` with all values redacted (showing only whether each was set and its character length), identified line 3 (`CENSUS_API_KEY=`) as the target, and gave two ways to edit it. Stated that the key should not be pasted into the chat, since the script reads it from disk.

Prompt 4

where where do i open it

Actions taken by AI: Opened `.env` in TextEdit for the user. Explained that the leading dot makes the file hidden in Finder and that `Cmd+Shift+.` toggles hidden files. No file contents were printed.

Prompt 5

ok i did it

Actions taken by AI: Verified the key was present without printing it (40 alphanumeric characters, the standard Census format), then ran `scripts/12_fetch_acs.py`.

ACS download completed. 25,764 county-year rows, 2015–2022; 3,220–3,222 counties per year. Raw API responses saved verbatim to `data/raw/acs/acs5_county_<year>.json`; derived file `data/processed/acs_county_2015_2022.csv`. Validation: `county_fips` is 5 characters throughout, no duplicate (year, county_fips), and missingness is negligible (1 county-year for per capita income, 38 for median gross rent, 13 for the "needs care" share).

Prompt 6

make sure it is hidden and create a venv if you have to

Actions taken by AI: 1. **Key-leak scan.** Read the key from `.env` and searched every other file in the project for that literal string — 100 files scanned (excluding `.git`, `venv` and `.env` itself). **Found in nothing.** Re-confirmed `.env` is gitignored, untracked and absent from all commits; `data/raw/` is gitignored; request URLs are logged with the key redacted. 2. **Virtual environment.** The existing `venv/` (Python 3.14) hangs when invoked from the Google Drive-synced folder. It was not needed: system Python 3.10.9 already provides pandas 1.5.3, numpy 1.23.5, scipy 1.10.0, matplotlib 3.7.0, statsmodels 0.14.0 and openpyxl 3.0.10. All scripts run on system Python; no new environment was created.

Problems 3–7

Prompt 7

ok after you finish that, here is your job: ok give me updates on each different question as you go through these with what i told you to do

now for question 3 use the crosswalk csv i just downloaded and merge that thru [year, oews_area] with the OEWS area wages down to counties. Make sure that you add leading zeros in, which is called zero pad. And add them to the OEWS area field and in the BLS data. And then we want to transform this into 7-character variable so that it matches a crosswalk frame format. So really making sure that the frameworks line up and then we can merge the data set effectively. Then we're going to attach the ACS county data that we found um, and label metros with area_title to match the OEWS crosswalks. Now all the data can be speaking the same terms as each other. And we should have a result that is given by year X county, and add them to the OEWS area field and in the BLS data. And then we want to transform this into seven character variable so that it matches a crosswalk frame format. So really making sure that the frameworks line up and that we can merge the data set effectively. Then we're going to attach the ACS county data that we found um, and label metros with to match the OEWS crosswalks. Now all the data can be speaking the same terms as each other. And we should have a result that is given by year X county. We also want to This is very important. Only have counties that are in an OEWS metropolitan area. Which means that oews_area starts with 00, this is very important, only have counties that are in an OEWS metropolitan area, which means that also as a sanity check, nearly all the counties should have wages. So if any are missing, we want to verify that the merge went correctly and that OEWS crosswalk was merged correctly with BLS, which may have a couple suppressions of area year cells. So maybe like 1% maximum missing. And this should all go under question three uh, as our data cleaning procedure and merging.

Okay, now for question four, here is, here's what we're going to do. You're going to create a descriptive table, which means you're going to display the mean, the standard deviation, and the number of observations for the predictors and the most recent year, which is 2022. Remember, the predictors that we had that I gave you in the ACS variables at the very beginning of this assignment.

Okay, now for question five. Here is what we are going to do. we are going to convert all of our dollar values to 2022 dollars, which is CPIU, because we need to make sure we're adjusting for inflation over time. And then we are going to create a plot with um, that puts childcare worker wages and rents together over time. So showing how in dollar values, childcare worker wages have changed over time and also showing how rents have changed over time in dollar value. There's likely going to be some kind of increase, at least nominally, because um, of inflation, but factoring out inflation is what we're really looking at here. What those trends look like for each of these variables over time. So that is question five.

Okay, now for question six. I want to make sure that I can do this by hand myself to check, and here's the information that I need. I need the mean of average wages adjusted for inflation across all years. I would love that displayed in a table. As well as the standard error for each average wage calculation each year. So like I want average wage of 2022 adjusted for inflation, and the standard error of the of the average wage adjusted for inflation in 2022. And do that for all of the years in the data set. And then from there, you can also calculate uh, a t-statistic and a p-value doing a hypothesis test to test the null that the average uh, inflation-adjusted wage in 2019 is, is equal to the average wage-adjusted wage with adjusted inflation in 2022. I want to do this by hand and test it out for myself, which is why I'm asking you for the table. But also run this just to confirm so I can compare them side by side and make sure that my hand-based calculation that I'm doing is matching what you're giving me, because I want to be able to understand this process and not just have you do it for me. [... the prompt then restated questions four, five and six a second time in near-identical wording ...]

Okay, and then for question seven, I want you to regress childcare worker wages on the share of women 25 plus with bachelor's degree or higher, which is a female college share variable. And only use a 2022 data. And once you run this regression, I just want to see the output table that you come up with for that. I'm going to calculate my test statistic and do my hypothesis test myself using the coefficients that you provide for me because I want to work through it myself.

Question 3 — Merge and cleaning

Script: `scripts/14_build_county_panel.py` **Output:** `data/processed/county_panel_2015_2022.csv` — 9,873 rows × 25 columns, one row per year × county.

Procedure: 1. Zero-padded both join keys to fixed-width strings **before** any merge: `county_fips` to 5 characters, `oews_area` to 7 characters, in the crosswalk, the OEWS extract and the ACS extract. Widths were asserted, not assumed. 2. Dropped 16 crosswalk rows carrying obsolete county FIPS — 12025 (Dade County FL, renamed 12086 Miami-Dade in 1997) and 51515 (Bedford city VA, merged into 51019 Bedford County in 2013). Both modern codes are already in the crosswalk for all eight years, so these were duplicates that could never match ACS; the script asserts the modern code is present before dropping. 3. Joined crosswalk to OEWS wages on `[year, oews_area]` (`validate="m:1"`). 4. Joined ACS county variables on `[year, county_fips]` (`validate="1:1"`). 5. Kept only counties whose `oews_area` begins 00 — the OEWS **metropolitan** sample.

Metropolitan-division backfill. The raw merge left 4.49% of metro county-years without a wage, concentrated at ~10% in 2015–2017 — the structural gap recorded under Problem 2. For those area-years only, divisions were aggregated to the parent MSA: `tot_emp` summed, `a_mean / h_mean` employment-weighted. Medians and percentiles were **not** aggregated — a pooled median cannot be recovered from division medians — and are left blank on those rows. Every backfilled row carries `wage_source = "division_aggregate"`; all others are `"published"`. 33 MSA-year cells were backfilled, covering 342 county-years. Mapping and values in `docs/oews_division_backfill.csv`.

Sanity check result: 99 of 9,873 metro county-years (1.00%) have no wage.

Year	Rows	Missing wage	%
2015	1,233	11	0.89
2016	1,233	14	1.14
2017	1,234	10	0.81
2018	1,233	5	0.41
2019	1,235	6	0.49
2020	1,235	20	1.62
2021	1,235	13	1.05
2022	1,235	20	1.62

The merge was verified rather than assumed: no duplicate (year, county_fips), row count unchanged by the ACS join, and both joins ran under pandas validate= cardinality checks. The residual 1.00% is BLS non-publication in small areas — not a merge failure. Per-county list in docs/q3_counties_missing_wages.csv.

Known remaining gap in ACS, disclosed: 7 county-years have no ACS predictors — the Connecticut counties in **2022 only**. The 2022 ACS 5-year replaced Connecticut's eight counties with nine planning regions (09110 – 09190), which do not map one-to-one onto the old county FIPS the crosswalk uses. No remap was attempted because the boundaries genuinely differ.

Question 4 — Descriptive statistics, 2022

Script: scripts/15_q4_descriptives.py · **Output:** docs/q4_descriptives_2022.md , output/q4_descriptives_2022.csv .
Sample: metro counties, 2022. Standard deviation is the sample SD (divisor n–1).

Predictor	Mean	Std. dev.	N
Per capita income (\$)	36,156.1	10,504.1	1,228
Total population	231,862.6	506,835.2	1,228
Own children under 6	15,440.9	34,275.0	1,228
Own children <6, all resident parents in LF	10,419.9	22,637.0	1,228
Median gross rent (\$/mo)	1,051.8	341.9	1,227
Share women 25+ with bachelor's+	0.3013	0.1070	1,228
Share women 16–64 full-time year-round	0.4367	0.0683	1,228
Share of population under 6	0.0660	0.0124	1,228
Share kids under 6 needing care	0.6728	0.0906	1,227

Outcome, for reference: childcare worker mean annual wage, mean 27,373.5, SD 4,337.1, N 1,215.

Question 5 — Constant 2022 dollars and the trend plot

Script: scripts/16_q5_real_trends.py · **Outputs:** output/q5_real_wages_rents.png , docs/q5_real_trends_table.md , data/processed/county_panel_real_2015_2022.csv .

CPI-U (series CUUR0000SA0 , annual averages) fetched from the BLS public API and cached to data/processed/cpi_u_annual.csv .
Deflation: $\text{real}_{2022} = \text{nominal}_t \times (\text{CPI}_{2022} / \text{CPI}_t)$. Applied to the childcare wage, median gross rent and per capita income.

Year	CPI-U	Deflator	Wage nominal	Wage real	Rent nominal	Rent real
2015	237.017	1.2347	20,937	25,851	805	993
2016	240.007	1.2194	21,462	26,170	821	1,001
2017	245.120	1.1939	22,106	26,393	847	1,011
2018	251.107	1.1655	22,743	26,506	876	1,020
2019	255.657	1.1447	23,490	26,889	900	1,030
2020	258.811	1.1308	24,445	27,642	920	1,040
2021	270.970	1.0800	25,598	27,646	968	1,045
2022	292.655	1.0000	27,373	27,373	1,052	1,052

Nominal 2015→2022: wage +30.74%, rent +30.72%. **Real** 2015→2022: wage +5.889%, rent +5.873%. The near-identical totals are a coincidence of the endpoints, not a coding error — the year-by-year paths differ (real wages rise to 2020–21 then fall 0.99% in 2022; real rent rises every year). This was checked explicitly.

The figure is three panels, not a dual-axis chart: real vs nominal wage; real vs nominal rent; and both real series indexed to 2015 = 100 on a common scale.

Question 6 — Mean real wage by year, and 2019 vs 2022

Script: `scripts/17_q6_wage_means_ttest.py` · **Output:** `docs/q6_wage_means_and_ttest.md` . $SE = sd / \sqrt{n}$, sample SD throughout.

County level, inflation-adjusted to 2022 dollars:

Year	N	Mean real wage	SD	SE
2015	1,222	25,851.21	3,266.89	93.4543
2016	1,219	26,169.71	3,371.54	96.5665
2017	1,224	26,392.52	3,514.92	100.4673
2018	1,228	26,505.90	3,702.79	105.6648
2019	1,229	26,888.87	3,897.24	111.1685
2020	1,215	27,641.46	4,423.70	126.9104
2021	1,222	27,646.21	4,464.55	127.7151
2022	1,215	27,373.48	4,337.10	124.4261

OEWS area level (one observation per area-year):

Year	N areas	Mean real wage	SD	SE
2015	381	25,937.65	3,270.72	167.5641
2016	381	26,278.38	3,323.28	170.2566
2017	384	26,510.27	3,475.07	177.3366
2018	385	26,710.36	3,727.65	189.9789
2019	385	27,189.98	4,021.61	204.9603
2020	376	28,126.27	4,579.23	236.1560
2021	382	28,041.27	4,553.03	232.9533
2022	377	27,716.31	4,409.76	227.1140

Welch two-sample t-test, H0: $\mu_{2019} = \mu_{2022}$, two-tailed, $\alpha = 0.05$ (county level)

```
diff      = 27,373.4815 - 26,888.8700 = 484.6115
SE(diff)  = sqrt(124.4261^2 + 111.1685^2) = 166.8541
t         = 484.6115 / 166.8541 = 2.9044
Welch df  = 2,408.506 ; t_crit(.975) = 1.9609
p         = 0.00371305
95% CI for the difference = [157.4191, 811.8039]
```

Cross-checked against `scipy.stats.ttest_ind(..., equal_var=False)`: identical. Pooled-variance alternative: $t = 2.9062$, $df = 2,442$, $p = 0.00369169$.

At the **area level** the same test gives $diff = 526.3327$, $SE = 305.9240$, $t = 1.7205$, $df = 750.450$, $p = 0.0857595$, $95\% CI = [-74.2360, 1,126.9014]$.

Dependence caveat recorded: OEWS publishes one wage per *area*, repeated across every county in that area, so county-level observations are not independent and county-level SEs are understated. Both levels are reported.

Question 7 — Wage on female college share, 2022

Script: `scripts/18_q7_regression.py` · **Output:** `docs/q7_regression_2022.md`

```
a_mean_i = b0 + b1 * share_women_25plus_bachelors_plus_i + e_i
```

OLS, 2022 only, metro counties, $N = 1,208$. 2022 is the CPI base year, so real and nominal wages coincide. The regressor is a 0–1 proportion, so $b1$ is the dollar change for a move from 0 to 100 percent; divide by 100 for a one-percentage-point change.

Term	Coefficient	Std. error	t	P> t	95% CI
share_women_25plus_bachelors_plus (b1)	16,499.9862	1,058.1894	15.5927	4.48293e-50	[14,423.8894, 18,576.0829]
Intercept (b0)	22,359.1942	339.1614	65.9249	<1e-300	[21,693.7823, 23,024.6060]

R-squared = 0.167777, Adjusted R-squared = 0.167087, Root MSE = 3,949.9201, $F(1, 1206) = 243.131$ ($p = 4.48293e-50$), residual $df = 1,206$.

Clustering by OEWS area (373 clusters) leaves the coefficients unchanged and roughly doubles the standard error: $b1 = 16,499.9862$, $SE = 2,217.3295$, $t = 7.4414$, $p = 6.97122e-13$.

Clarification and follow-ups

Prompt 8

Maybe to clarify for six, let me tell you the test that I'm running. It's assessing whether or not average inflation-adjusted wage earnings for childcare workers are the same in 22 as in 2019. That's what I'm looking for. That's what I'm testing. So I want to see if the means of those values are different and use the standard errors from each of those years.

Actions taken by AI: Confirmed the specification matches the test already run in Question 6 — a two-sample difference of means on inflation-adjusted wages, with $SE(diff) = \sqrt{SE_{2019}^2 + SE_{2022}^2}$. No recomputation was required and no previously reported number changed.

Additionally checked the sample structure: 1,212 of the 1,215 counties with a 2022 wage (99.8%) also appear in 2019, so the two years are near-identical sets of counties rather than independent samples. A **paired** test was run as a supplement, using $SE = \frac{sd(x_{2022} - x_{2019})}{\sqrt{n}}$:

Specification	diff	SE	t	df	p
Two-sample, county level (as specified)	484.6115	166.8541	2.9044	2,408.5	0.003713
Two-sample, area level	526.3327	305.9240	1.7205	750.5	0.085760
Paired, county level	456.3581	53.2066	8.5771	1,211	2.93e-17
Paired, area level	497.1583	111.5536	4.4567	373	1.10e-05

The test as the user specified it remains the headline; both are retained in `docs/q6_wage_means_and_ttest.md`.

Prompt 9

give me the record of the prompts now and this as a writeup with the charts and data and headers i asked for BUT NO WRITING THAT YOU CREATED -- JUST THE NUMBERS AND TABLES, NO INTERPRETATION OR AI GENERATED WRITING

Actions taken by AI: Wrote `scripts/19_build_writeup.py`, which regenerates the write-up from the processed data files at build time so the document cannot drift from the data. Produced `docs/WRITEUP.md` and a self-contained `docs/WRITEUP.html` with the figure embedded. Results sections contain only headers, tables, the figure and the formulas defining each computation. Reject / fail-to-reject decisions were deliberately omitted so the user performs the hypothesis tests.

Prompt 10

ok make these downloadable and put them all in pdf with the charts/tables and specific thigns i asked you for results for first and then follow up with prompts. remmeber, no AI GENERATED WRITING, I CANNOT SUBMIT THAT, JUST THE NUMBERS AND FIGURES YOU CREATE. FOR QUESTIONS LIKE 1 WHER EYOU JUST DOWNLOAD OR MERGE DATA, SAY "Process completed. See AI output notes below."

Actions taken by AI: Rebuilt the document: Questions 1 and 2 reduced to the single line `Process completed. See AI output notes below.`; Question 3 the same line plus the missingness table the user had requested as a sanity check; Questions 4–7 unchanged tables and figure. Narrative preamble removed. The prompt record was appended as an "AI output notes" section. Rendered to `output/Ventresca_writeup.pdf` (19 pages) with headless Chrome, as the machine has no pandoc, LaTeX or reportlab.

Prompt 11

why is there 1.82 percent in the merge missing

Actions taken by AI: Searched the generated documents — no 1.82% value exists; the maximum is 1.62% (2020 and 2022) and the overall rate is 1.00%. Diagnosed the cause of the missing cells:

Cause	Count
BLS published no 39-9011 row at all for that area-year	95
Row exists but <code>A_MEAN</code> suppressed (flag <code>*</code>)	4

Puerto Rico accounts for 36 of the 99 missing cells (6.5% of PR county-years, versus 0.68% elsewhere). Excluding Puerto Rico the overall rate falls to 0.68%. No change was made to the sample; the option was put to the user, who chose to keep it as is.

Prompt 12

no it is fine, can you just give me a link for someone to be able to reproduce this

Actions taken by AI: Verified that nothing secret would be published (no `.env`, no key in any tracked file, `data/raw/` and `venv/` excluded) and asked whether to publish. The user declined at that point; nothing was pushed.

Prompt 13

push to new repo github hw 5 and push

Actions taken by AI: Created the public repository `github.com/emmaventresca/hw5` and pushed. Added a "Reproduce" section to the README (dependencies, run order for scripts 12–19, note that the reader must supply their own free Census API key) and a `.env.example` template containing no key. Re-verified after pushing that the remote tree contains no `.env` , no `data/raw/` , no `venv/` , and that the key appears in no pushed blob.

Prompt 14

wait no can you redo the repo and writeup so the prompts literally just start where this chat started today like hey ok udner hw five folder thing i told you, that is where i started. the other stuff was from a failed previous attempt, plz disregard, just this chaat@

Actions taken by AI: Rewrote this prompt log so that it contains only the prompts from this session, beginning at Prompt 1 above. Rebuilt `docs/WRITEUP.md` , `docs/WRITEUP.html` and `output/Ventresca_writeup.pdf` from the revised log, and rebuilt the GitHub repository with a single clean commit containing only this session's work.

Data integrity (SHA-256)

<code>data/processed/acs_county_2015_2022.csv</code>	<code>d25b81341cd4db16ec788ea883a4a4c5b538deb88b7f92d1ce722ab52748844c</code>
<code>data/processed/oews_childcare_2015_2022.csv</code>	<code>5d89cd9d954a653d5ab24fa7c687db15ab26e00c06559ace9a38eb54d37aea61</code>
<code>data/processed/county_panel_2015_2022.csv</code>	<code>a013c6975daf6be155cdf208b68ebf2bacee1714dd8bd37a9888b0549a2f7464</code>
<code>data/processed/county_panel_real_2015_2022.csv</code>	<code>8ef0b98cedd6ec3d11d40a282014e016e4b05e88eed0919cb55f55799e3336cd</code>
<code>data/processed/cpi_u_annual.csv</code>	<code>2437bf71c55963d4cc23d37f8f556f115428614824dbe8dd12ca5fbab7ee60fc</code>
<code>data/crosswalks/county_oews_crosswalk.csv</code>	<code>5df fe4ce5eb782e9b56db6bdac61153fa18848dff4fe3d82b13b011aaeb196da</code>

Raw ACS JSON and OEWS archives in `data/raw/` are the files as downloaded and were never edited.